



The Ephemeris



December 2018

Volume 29 Number 04- The Official Publication of the San José Astronomical Association

December 2018 - March 2019 Events

Board & General Meetings

Saturday 12/15, 1/19, 2/16, 3/23

Board Meetings: 6 -7:30pm

General Meetings: 7:30-9:30pm

Fix-It Day (2-4pm)

Sunday 12/2, 1/6, 2/3, 3/3

Solar Sunday (1:30-3:30pm)

Sunday 12/2, 1/6, 2/3, 3/3

Intro to the Night Sky Class & Houge Park 1Q In-Town Star Party

Friday 12/14, 1/11, 2/15, 3/15

Houge Park 3Q In-Town Star Party

Friday 12/28, 1/25, 2/22, 3/29

RCDO Starry Nights Star Party

Saturday 12/29, 1/26, 2/23, 3/30

Imaging SIG Mtg

Tuesday 12/18, 1/15, 2/19, 3/19

Astro Imaging Clinics (at Coyote Valley OSP)

Saturday 3/9

Quick START

Quarterly (by appointment)

Binocular Star Gazing

Will resume in Summer 2019

Please refer to the SJAA Website home page (www.sjaa.net) and click on any "Upcoming Event" highlighted in Blue. It will take you to the SJ-Astronomy Meetup page (www.meetup.com/SJ-Astronomy/events/) where you can find specific event times, locations, maps and possible cancellation due to weather.

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Photo by SJAA Member: Alan Pham
September 10, 2018
Location: San Jose, California



Fall Swap Meet

Dave Ittner



SJAA held its annual Fall Swap Meet on Sunday, Oct. 28, 2018 at the Houge Park Hall. There was a large queue of buyers waiting for the doors to open at noon on Sunday.

Thank you all for making the trip to San Jose and making our Swap Meet a success. SJAA brought in nearly \$2,000.

The next Swap Meet is tentatively set for March 17, 2019. Please watch the SJAA website for additional information as the date gets closer.

SJAA wishes its members Happy Holidays and Best Wishes for a Happy New Year!



Welcome New Directors: Ken Miura and Sandy Mohan

Dear SJAA members, I would like to introduce the new members of the SJAA Board of Directors, Ken Miura and Sandy Mohan. They have stepped up to fill the positions vacated by Teruo Utsumi and Bill O'Neil respectively, and they will finish out the remaining terms of the departing board members. Included below are brief self-introductions from Ken and Sandy.

Ken & Sandy, congratulations and welcome to the SJAA Board of Directors.

Swami Nigam
President

Ken Miura



I am a computer scientist, specialized in supercomputing. I first became a SJAA member in 1988, while I was working for Fujitsu America Inc. until I went back to Japan in 1997. Now I am retired and back to the Bay area again since 2014. My telescopes are mostly

Takahashi fluorite refractors; I usually bring FS128 to the star parties. Takahashi TOA150 is too heavy to bring here, unfortunately, so it is still sitting in the closet of my apartment in Tokyo. I also serve as a registered volunteer at the Onizuka Center (Mauna Kea) in Hawaii Island, although I don't go there very often.

Ken Miura
Director

Sandy Mohan



I'm honored to be elected to the Board of Directors of SJAA. I've been a member of the SJAA since 2000 and served as the Content Editor for Ephemeris for the last 5 years. I live in Saratoga with my husband. I have two grown sons who originally got me interested in astronomy when they were little. I used to bring them regularly to the Star Parties at Hough Park when we moved from NJ to the Bay Area almost 18 years ago. These days, I'm perfectly content sitting in my backyard with a telescope and scanning the skies for familiar and new objects the old-fashioned way. And when my sons visit us, I make it a point to take them on a tour of the night sky. I'm a molecular biologist by training (PhD from NYU) and work for a biotech company in South San Francisco.

Sandy Mohan
Director

Instagram and Social Media

Jessica Johnson

Are you interested in SJAA events and astronomy news? Join us on Instagram @sjastronomy to follow pictures of our events, announcements for SJAA and astronomy news from NASA, ESA, ISS and more! You can use Instagram for free as an app or on any web browser to keep up with our posts.



Isaac Kikawada Farewell

By Heidi Gerster



Isaac Kikawada

Photograph courtesy: Heidi Gerster



Isaac Kikawada with his wife Heidi Gerster in Hawaii in 2012

Photograph courtesy: Heidi Gerster

My husband and long-time member of SJAA, Isaac Kikawada, died in July 2018. He was born in Sendai, Japan and came to the US by boat in 1956 to attend Kenyon College and earn his bachelor's degree. He continued and finished his studies in the Department of Near Eastern Studies at UC Berkeley where he taught the languages and literature of the Ancient Near East until his retirement.

After moving to Mountain View, he became more active in amateur astronomy. Isaac loved teaching others, especially young people. This drew him to "Project Astro" which paired him with the science teacher at Blach School for teaching astronomy. You may remember volunteering at his moon, sun and star parties on the school tennis courts. Your help made these events very memorable and encouraged students to look up and follow events in the sky. Isaac loved observing in dark skies, but also did a lot of backyard astronomy. He was very interested in the movement of planets, loved the beauty of double stars, fuzzy objects, and made up names for asterisms he discovered. Solar and lunar eclipses and the Northern Lights were special treats and brought us to places we would otherwise never have visited. Last year we took a road trip to Casper, WY, for the "All American Eclipse of 2017". Isaac wondered on the way home if we would be around for the 2024 Eclipse. I will be on my own for that trip, but his memory will be with all of us.



Isaac Kikawada with his telescope in Switzerland in 2004

Photograph courtesy: Heidi Gerster



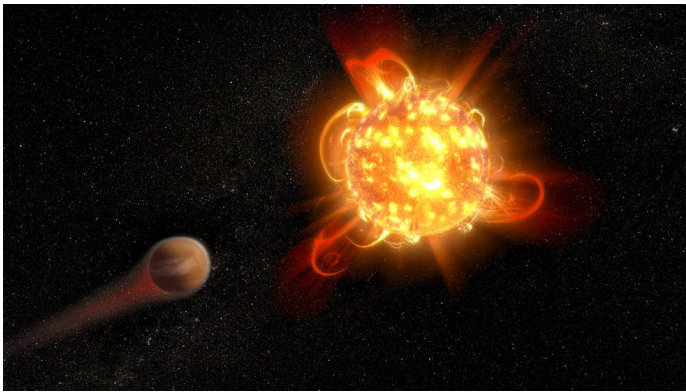
Superflares From Young Red Dwarf Stars Imperil Planets

Credit: NASA

The word "HAZMAT" describes substances that pose a risk to the environment, or even to life itself. Imagine the term being applied to entire planets, where violent flares from the host star may make worlds uninhabitable by affecting their atmospheres.

NASA's Hubble Space Telescope is observing such stars through a large program called HAZMAT — Habitable Zones and M dwarf Activity across Time.

"M dwarf" is the astronomical term for a red dwarf star — the smallest, most abundant and longest-lived type of star in our galaxy. The HAZMAT program is an ultraviolet survey of red dwarfs at three different ages: young, intermediate and old.



Violent outbursts of seething gas from young red dwarf stars may make conditions uninhabitable on fledgling planets. In this artist's rendering, an active, young red dwarf (right) is stripping the atmosphere from an orbiting planet (left). Scientists found that flares from the youngest red dwarfs they surveyed — approximately 40 million years old — are 100 to 1,000 times more energetic than when the stars are older. They also detected one of the most intense stellar flares ever observed in ultraviolet light — more energetic than the most powerful flare ever recorded from our Sun.

Credits: NASA, ESA and D. Player (STScI)

Stellar flares from red dwarfs are particularly bright in ultraviolet wavelengths, compared with Sun-like stars. Hubble's ultraviolet sensitivity makes the telescope very valuable for observing these flares. The flares are believed to be powered by intense magnetic fields that get tangled by the roiling motions of the stellar atmosphere. When the tangling gets too intense, the fields break and reconnect, unleashing tremendous amounts of energy.

The team has found that the flares from the youngest red dwarfs they surveyed — just about 40 million years old — are 100 to 1,000 times more energetic than when the stars are older. This younger age is when terrestrial planets are forming around their stars.

Approximately three-quarters of the stars in our galaxy are red dwarfs. Most of the galaxy's "habitable-zone" planets — planets orbiting their stars at a distance where temperatures are moderate enough for liquid water to exist on their surface — likely orbit red dwarfs. In fact, the nearest star to

our Sun, a red dwarf named Proxima Centauri, has an Earth-size planet in its habitable zone.

However, young red dwarfs are active stars, producing ultraviolet flares that blast out so much energy that they could influence atmospheric chemistry and possibly strip off the atmospheres of these fledgling planets.

"The goal of the HAZMAT program is to help understand the habitability of planets around low-mass stars," explained Arizona State University's Evgenya Shkolnik, the program's principal investigator. "These low-mass stars are critically important in understanding planetary atmospheres."

The results of the first part of this Hubble program are being published in *The Astrophysical Journal*. This study examines the flare frequency of 12 young red dwarfs. "Getting these data on the young stars has been especially important, because the difference in their flare activity is quite large as compared to older stars," said Arizona State University's Parke Loyd, the first author on this paper.

The observing program detected one of the most intense stellar flares ever observed in ultraviolet light. Dubbed the "Hazflare," this event was more energetic than the most powerful flare from our Sun ever recorded.

"With the Sun, we have a hundred years of good observations," Loyd said. "And in that time, we've seen one, maybe two, flares that have an energy approaching that of the Hazflare. In a little less than a day's worth of Hubble observations of these young stars, we caught the Hazflare, which means that we're looking at superflares happening every day or even a few times a day."

Could super-flares of such frequency and intensity bathe young planets in so much ultraviolet radiation that they forever doom chances of habitability? According to Loyd, "Flares like we observed have the capacity to strip away the atmosphere from a planet. But that doesn't necessarily mean doom and gloom for life on the planet. It just might be different life than we imagine. Or there might be other processes that could replenish the atmosphere of the planet. It's certainly a harsh environment, but I would hesitate to say that it is a sterile environment."

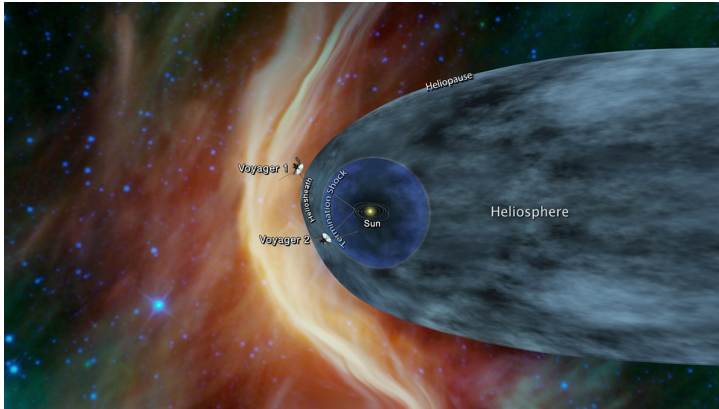
The next part of the HAZMAT study will be to study intermediate-aged red dwarfs that are 650 million years old. Then the oldest red dwarfs will be analyzed and compared with the young and intermediate stars to understand the evolution of the ultraviolet radiation environment of low-mass planets around these low-mass stars.

The Hubble Space Telescope is a project of international cooperation between NASA and ESA (European Space Agency). NASA's Goddard Space Flight Center in Greenbelt, Maryland, manages the telescope. The Space Telescope Science Institute (STScI) in Baltimore, Maryland, conducts Hubble science operations. STScI is operated for NASA by the Association of Universities for Research in Astronomy, in Washington, D.C.



NASA Voyager 2 Could Be Nearing Interstellar Space

Credit: NASA JPL



This graphic shows the position of the Voyager 1 and Voyager 2 probes relative to the heliosphere, a protective bubble created by the Sun that extends well past the orbit of Pluto. Voyager 1 crossed the heliopause, or the edge of the heliosphere, in 2012. Voyager 2 is still in the heliosheath, or the outermost part of the heliosphere. Image Credit: NASA/JPL-Caltech

NASA's Voyager 2 probe, currently on a journey toward interstellar space, has detected an increase in cosmic rays that originate outside our solar system. Launched in 1977, Voyager 2 is a little less than 11 billion miles (about 17.7 billion kilometers) from Earth, or more than 118 times the distance from Earth to the Sun.

Since 2007 the probe has been traveling through the outermost layer of the heliosphere -- the vast bubble around the Sun and the planets dominated by solar material and magnetic fields. Voyager scientists have been watching for the spacecraft to reach the outer boundary of the heliosphere, known as the heliopause. Once Voyager 2 exits the heliosphere, it will become the second human-made object, after Voyager 1, to enter interstellar space.

Since late August, the Cosmic Ray Subsystem instrument on Voyager 2 has measured about a 5 percent increase in the rate of cosmic rays hitting the spacecraft compared to early August. The probe's Low-Energy Charged Particle instrument has detected a similar increase in higher-energy cosmic rays.

Cosmic rays are fast-moving particles that originate outside the solar system. Some of these cosmic rays are blocked by the heliosphere, so mission planners expect that Voyager 2 will measure an increase in the rate of cosmic rays as it approaches and crosses the boundary of the heliosphere.

In May 2012, Voyager 1 experienced an increase in the rate of cosmic rays similar to what Voyager 2 is now detecting. That was about three months before Voyager 1 crossed the heliopause and entered interstellar space.

However, Voyager team members note that the increase in cosmic rays is not a definitive sign that the probe is about to cross the heliopause. Voyager 2 is in a different location in the heliosheath -- the outer region of the heliosphere -- than Voyager 1 had been, and possible differences in these locations means Voyager 2 may experience a different exit timeline than Voyager 1.

The fact that Voyager 2 may be approaching the heliopause six years after Voyager 1 is also relevant, because the heliopause moves inward and outward during the Sun's 11-year activity cycle. Solar activity refers to emissions from the Sun, including solar flares and eruptions of material called coronal mass ejections. During the 11-year solar cycle, the Sun reaches both a maximum and a minimum level of activity.

"We're seeing a change in the environment around Voyager 2, there's no doubt about that," said Voyager Project Scientist Ed Stone, based at Caltech in Pasadena. "We're going to learn a lot in the coming months, but we still don't know when we'll reach the heliopause. We're not there yet -- that's one thing I can say with confidence."

The Voyager spacecraft were built by NASA's Jet Propulsion Laboratory in Pasadena, California, which continues to operate both. JPL is a division of Caltech. The Voyager missions are a part of the NASA Heliophysics System Observatory, managed by the Heliophysics Division of the Science Mission Directorate in Washington.

For more information about the Voyager spacecraft, visit:

<https://www.nasa.gov/voyager>



European-Built Service Module Arrives in U.S. for First Orion Moon Mission

Credit: NASA



The European Service Module for NASA's Orion spacecraft is loaded on an Antonov airplane in Bremen, Germany, on Nov. 5, 2018, for transport to NASA's Kennedy Space Center in Florida. For the first time, NASA will use a European-built system as a critical element to power an American spacecraft, extending the international cooperation of the International Space Station into deep space.

Credits: NASA/Rad Sinyak

The powerhouse that will help NASA's Orion spacecraft venture beyond the Moon is stateside. The European-built service module that will propel, power and cool during Orion flight to the Moon on Exploration Mission-1 arrived from Germany at the agency's Kennedy Space Center in Florida on Tuesday to begin final outfitting, integration and testing with the crew module and other Orion elements.

The service module is integral to human missions to the Moon and Mars. After Orion launches on top of the agency's Space Launch System rocket, the service module will be responsible for in-space maneuvering throughout the mission, including course corrections. The service module will also provide the powerful burns to insert Orion into lunar orbit and again to get out of lunar orbit and return to Earth. It is provided by ESA (European Space Agency) and built by ESA's prime contractor Airbus of Bremen, Germany. NASA's prime contractor for Orion, Lockheed Martin, built the crew module and other elements of the spacecraft.

"We have a strong foundation of cooperation with ESA through the International Space Station partnership, and the arrival of the service module signifies that our international collaboration extends to our deep space human exploration efforts as well," said Bill Gerstenmaier, NASA's associate administrator for Human Exploration and Operations.

The European-built service module brings together new technology and lightweight materials while taking advantage of spaceflight-proven hardware. It is comprised of

more than 20,000 components, including four solar array wings that provide enough electricity to power two three-bedroom homes, as well as an orbital maneuvering system engine, a recently refurbished engine previously used for in-orbit control by the space shuttle. Beginning with Exploration Mission-2, the module also will provide air and water for astronauts flying inside Orion, which will carry people to destinations farther than anyone has travelled before and return them safely to Earth.

"Our teams have worked together incredibly hard to develop a service module that will make missions to the Moon and beyond a reality," said Mark Kirasich, NASA's Orion program manager. "It is quite an accomplishment of ESA and Airbus to have completed the developmental work on the module and have this major delivery milestone behind us."

Now that the service module is at Kennedy, it will undergo a host of tests and integration work ahead of Exploration Mission-1. Engineers will complete functional checkouts to ensure all elements are working properly before it is connected to the Orion crew module. Teams will weld together fluid lines to route gases and fuel and make electrical wiring connections. The service module and crew module will be mated, and the combined spacecraft will be sent to NASA's Glenn Research Center's Plum Brook Station in Ohio early next year where it will undergo 60 days of continuous testing in the world's largest thermal vacuum chamber to ensure Orion can withstand the harsh environment of deep space. Once that testing is complete, it will return to Kennedy for integration with the SLS rocket in preparation for launch.

NASA is leading the next steps to establish a permanent human presence at the Moon. The first in a series of increasingly complex missions, Exploration Mission-1 is a flight test of an uncrewed Orion spacecraft and SLS rocket that will launch from NASA's modernized spaceport at Kennedy. The mission will send Orion 40,000 miles beyond the Moon and back and pave the road for future missions with astronauts. Together, NASA and its partners will build the infrastructure needed to explore the Moon for decades to come while laying the groundwork for future missions to Mars.



Thor's Helmet
Glenn Newell



Date: Sep 30, 2018

Location: Siding Springs, Australia.

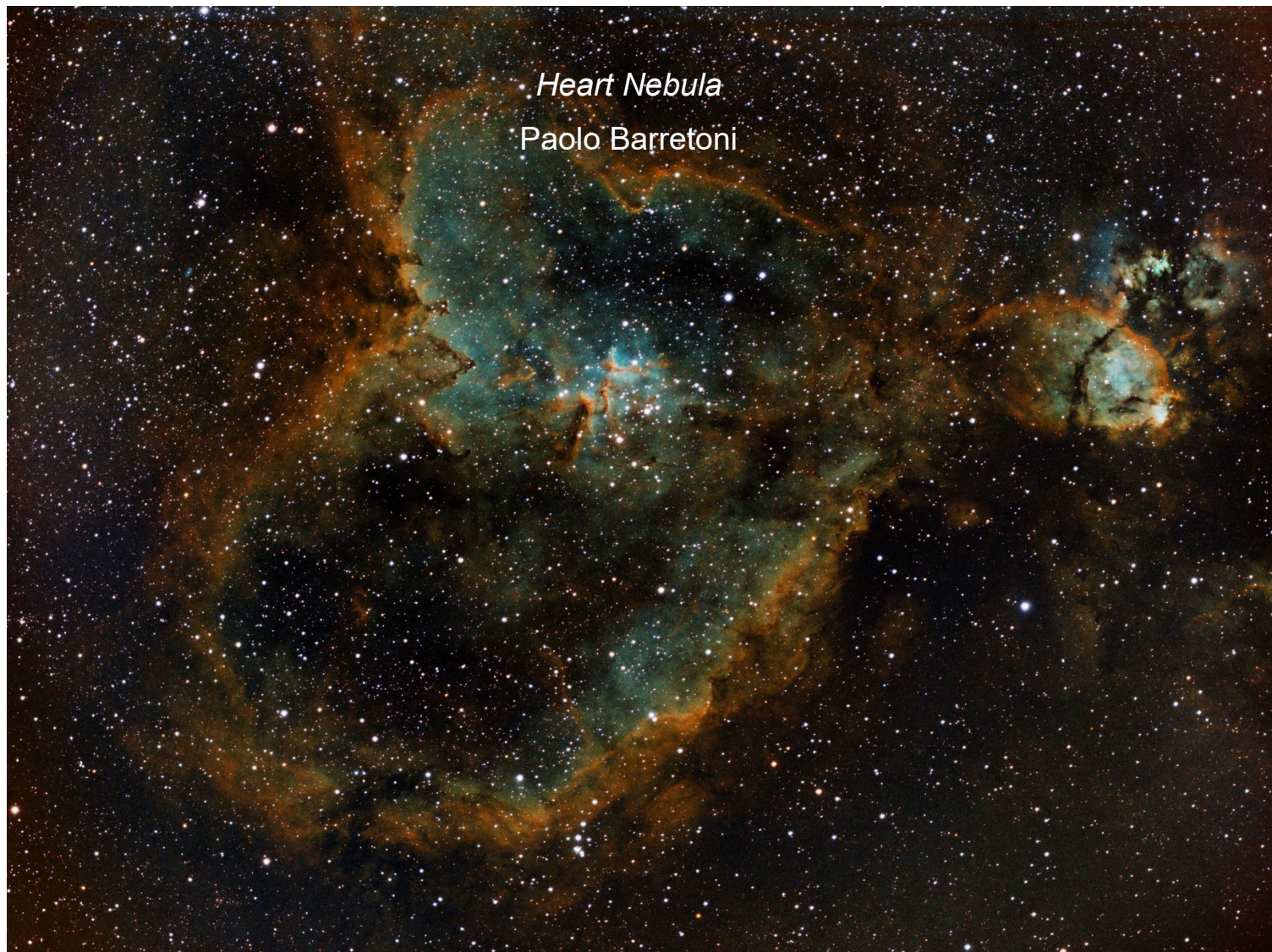
Telescope: Planewave Instruments Planewave 20" CDK

Camera: FLI PL16803

Mount: Planewave Ascensio 200HR

Software: Adobe PS CC, Topaz Labs DeNoise, Picasa, CCDStack2





Dates: Oct. 19, 2018; Oct. 20, 2018; Oct 21, 2018; Oct. 23, 2018; Oct. 26, 2018; Oct. 29, 2018

Location: San Jose, CA

Telescope: Canon 400mm

Camera: Orion Star Shoot AutoGuider

Mount: Orion Atlas AZ/EQ Pro

Software: AstroPixel Processor, MaximDL, Photoshop CS6



SJAA Member Astrophoto Gallery



Date: Sep 7, 2018

Location: N/A

Telescope: TS Optics 8" f/4 ONTC Newton

Camera: ZWO ASI120 MM

Mount: Orion Atlas AZ/EQ Pro

Software: Pleiades Astrophoto PixInsight 1.8 Ripley



Kid Spot



Kid Spot Jokes:

- ♦ **What kind of tunes do planets sing?**
Nep-tunes.
- ♦ **What kind of light goes around the earth?**
Satel-lite.
- ♦ **What do you call crazy moon?**
Luna-tic

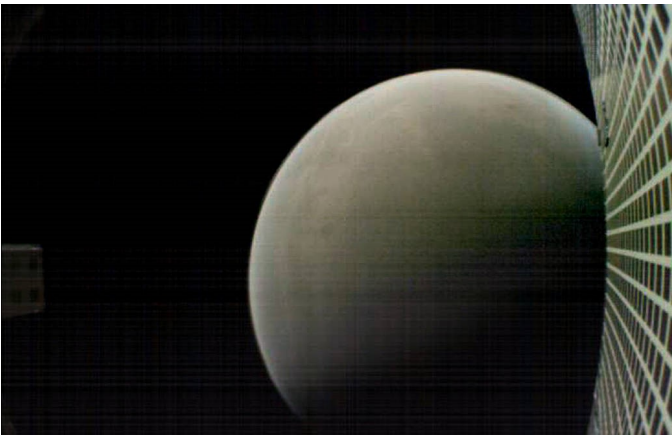
Kid Spot Quiz:

1. **The Pillars of Creation as seen in the Hubble Space Telescope is part of which nebula?**
a) Trifid Nebula b) Orion Nebula c) Eagle Nebula
2. **In which constellation is the double star Alberio?**
a) Cygnus b) Gemini c) Taurus
3. **By what colloquial name is Pleiades Cluster, M45 also known?**
a) Seven Princes b) Seven Sisters c) Shepherd's Bush



Farewell to Mars

Credit: NASA/JPL-Caltech



4,700 miles (7,600 kilometers) away during its flyby of the Red Planet on Nov. 26, 2018. MarCO-B was flying by Mars with its twin, MarCO-A, to attempt to serve as communications relays for NASA's InSight spacecraft as it landed on Mars. This image was taken at about 12:10 p.m. PST (3:10 p.m. EST) while MarCO-B was flying away from the planet after InSight landed.

The MarCO and InSight projects are managed for NASA's Science Mission Directorate, Washington, by JPL, a division of the California Institute of Technology, Pasadena.

MarCO-B, one of the experimental Mars Cube One (MarCO) CubeSats, took this image of Mars from about

Kid Spot Quiz Answers:

1. C
2. A
3. B



From the Board of Directors

Announcements

- ◇ Sandy Mohan is appointed to the Board of Directors
- ◇ SJAA Spring Swap Meet is tentatively scheduled for March 17, 2019, 12 noon.
- ◇ Dave Ittner will step down from Membership Chair and Donations Chair positions at the end of December

Board Meeting Excerpts

August 25, 2018

In attendance

Swami Nigam, Rob Jaworski, Glenn Newell, Ken Miura, Wolf Witt, Vini Carter, Rob Chapman

Excused Absences: Sukhada Palav
Guest: Emi Nikolov, Gerry, Gary, Jessica Johnson

Open Board Seat

A seat on the Board of Directors has opened up due to the resignation of Bill O'Neil. The remaining term for this Director seat is until February 2020. Swami is considering proposing Sandy Mohan to fill the seat for the remainder of the term.

Open Vice President Position

The position of Vice President has opened up due to the resignation of Bill O'Neil. The remaining term for this position is until March 2019. Swami would like to temporarily fill this position until the next officer elections with someone who has been actively involved with SJAA.

Open Content Editor Position

Sandy Mohan has decided to step down from this position. Emi Nikolov will cover for this position until a replacement is found.

2019 Annual Calendar

Rob Chapman and Dave Ittner are working on the 2019 club calendar

Starry Nights with OSA

Swami Nigam met with Teri Rogoway of OSA to discuss various ideas about the SJAA programs run jointly with OSA.

September 22, 2018

In attendance

Swami Nigam, Sukhada Palav, Wolf Witt, Vini Carter, Rob Jaworski, Glenn Newell

Excused Absences: Rob Chapman, Ken Miura

Guest: Emi Nikolov, Rashi, Bob, Natti, Cat, Marianne, Gary

Mendoza Ranch Observing Program

The board decided to make the monthly Mendoza Ranch observing event a formal SJAA program. Natti and Bob Thompson are the official leaders of this new program, and can be reached at MendozaRanch@sjaa.net.

Secure Storage for Houge Park

The current storage at Houge Park needs better organizing and security. Kanch and Dave have been requested to come up with a plan for securing SJAA assets stored in that area.

Star Party Request

We get many requests for Star Parties, often from commercial organizations. SJAA is happy to participate in various types of educational and outreach events organized by various organizations. The board feels that SJAA should not participate in commercial and/or for-free events. A formal policy and form letter response will be drafted soon.

October 27, 2018

In attendance

Swami Nigam, Rob Jaworski, Wolf Witt, Vini Carter, Sukhada Palav, Glenn Newell

Excused Absences: Rob Chapman, Ken Miura

Guest: Emi, Nikola, Victor, Krassi Antonova (of the Nikolov family), Jessica Johnson, Sandy Mohan, Kunch, Marianne Damon

Empty Board Seat

Sandy Mohan was appointed by the board to fill the open position on the Board of Directors, until the end of this position's term in February 2020.

Ephemeris Content Editor

Joe Maille expressed an interest in taking up the Ephemeris Content Editor position but decided not to.

Recruiting apprentices for all leaders/officers; job descriptions

In an effort to recruit more SJAA members to help out with the club operations and provide backup to the program leaders, all leaders and officers will be requested to write up a brief job description.

Dave Ittner to vacate his positions in December

Dave Ittner would like to step down from the Membership Chair and Donations Chair positions at the end of this year. Search is on for replacements. He would also like to hand over the responsibilities of running the sjaaobservers Yahoo group and the SJ Astronomy meetup to someone else.

Secure Storage for Houge Park

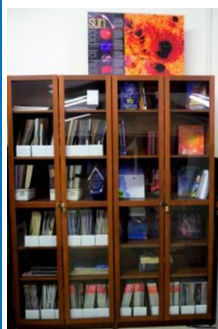
Kanch presented a proposal to better organize and secure club assets at Houge Park.

2019 Annual Calendar Update

Tentative 2019 calendar is ready and currently being reviewed by the board.



SJAA Library



SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers! You may still send all your questions/comments

to librarian@sjaa.net.

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. Headed up by Vini Carter, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

<http://www.sjaa.net/programs/fixit/>

Solar Observing

Solar observing sessions, headed up by Wolf Witt, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change.

<http://www.meetup.com/SJ-Astronomy/>

Quick STArT Program

The Quick STArT Program, headed up by Dave Ittner, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick Start sessions are generally held quarterly. The next Quick Start session during this time period is October 12th.

<http://www.sjaa.net/programs/quick-start/>

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, run by David Grover, each with a similar format, with only the content chang-

ing to reflect what's currently in the night sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with views through SJAA Telescopes at the In-Town Star Party, to get a better look at the night's celestial objects.

<http://www.sjaa.net/programs/beginners-astronomy/>

Loaner Program

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met. If you are an SJAA member and an experienced observer or have been through the SJAA Quick STArT program please fill this form to request a particular item. Please also consider donating unused equipment.

<http://www.sjaa.net/programs/loaner-telescope-program/>

Astro Imaging Special Interest Group (SIG)

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. Run by Bruce Braunstein, the Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

<http://www.sjaa.net/programs/imaging-sig/>

Astro Imaging Workshops and Field Clinics

Not to be confused with the SIG group this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astrophotography, to have a chance of seeing what it is all about.

Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's (Program Coordinator) for additional information.

<http://www.sjaa.net/programs/school-star-party/>

SJAA Contacts

President/Dir:	Swami Nigam
Vice President:	Vacant
Treasurer/Dir:	Rob Jaworski
Secretary/Dir:	Rob Chapman
Director:	Sandy Mohan
Director:	Vini Carter
Director:	Ken Miura
Director:	Glenn Newell
Director:	Wolf Witt
Director:	Sukhada Palav
Ephemeris Newsletter -	
Editor:	Sandy Mohan
Prod. Editor:	Emi Nikolov
Binocular Stargazing:	Ed Wong
Fix-it Program:	Vini Carter
Hands on Imaging:	Glenn Newell
Imaging SIG:	Bruce Braunstein
Intro to the Night Sky:	David Grover
Library:	Jai Purandare
Loaner Program:	Muditha Kanchana
Memberships:	Dave Ittner*
Publicity:	Rob Jaworski
Quick STArT	Dave Ittner
Donations:	Dave Ittner*
Solar:	Wolf Witt
Starry Nights:	Carl Wong
School Events:	Cat Cvengros
Refreshments:	Marianne Damon
Social Manager:	Jessica Johnson
Speakers:	Sukhada Palav
Webmaster:	Satish Vellanki
E-mails:	http://www.sjaa.net/contact

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles, including Observation Reports for publication should be submitted by no later than the 20th of the month of February, May, August and November.

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243
<http://www.sjaa.net/contact>

Place
postage
here

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Fold here

San Jose Astronomical Association Annual Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

Membership Type (must be 18 years or older):

☐ **New** ☐ **Renewal**

☐ **\$20** Regular Membership with online Ephemeris

☐ **\$30** Regular Membership with hardcopy Ephemeris mailed to below address

The newsletter is always available online at: <http://www.sjaa.net/sjaa-newsletter-ephemeris/>

Questions? Send e-mail to: memberships@sjaa.net

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